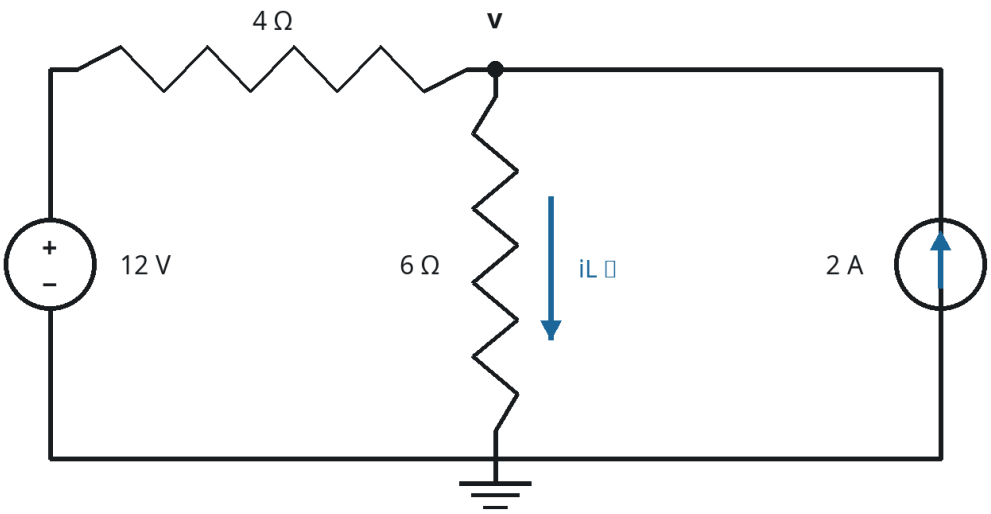
Superposition Questions

Question 1

Use superposition to find the downward current in the $6\,\Omega$ resistor.

One-node superposition circuit



Ground is the bottom wire. A grounded 12 V source feeds node v through 4 Ω. A 6 Ω resistor joins v to ground with downward load current. A 2 A source points upward from ground to v.

Solution

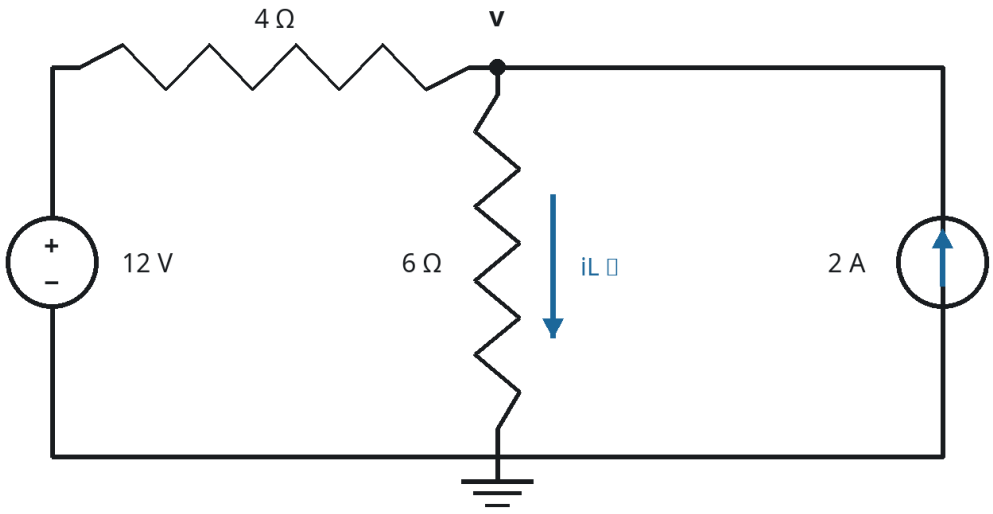
Take ground as zero and downward load current as positive. With the current source open, $v_V = 12\left(\frac{6}{4+6}\right) = 7.20\,\text{V}, \qquad i_{6,V} = \frac{7.20}{6} = 1.20\,\text{A}$

With the voltage source shorted, $v_I = 2(4 \parallel 6) = 2(2.40) = 4.80\,\text{V}, \qquad i_{6,I} = \frac{4.80}{6} = 0.800\,\text{A}$ $i_6 = i_{6,V} + i_{6,I} = 2.00\,\text{A}$ downward

Question 2

For the same circuit, find the node voltage and power absorbed by the $6\,\Omega$ resistor. Explain why power is not directly superposed.

One-node superposition circuit



Ground is the bottom wire. A grounded 12 V source feeds node v through 4 Ω. A 6 Ω resistor joins v to ground with downward load current. A 2 A source points upward from ground to v.

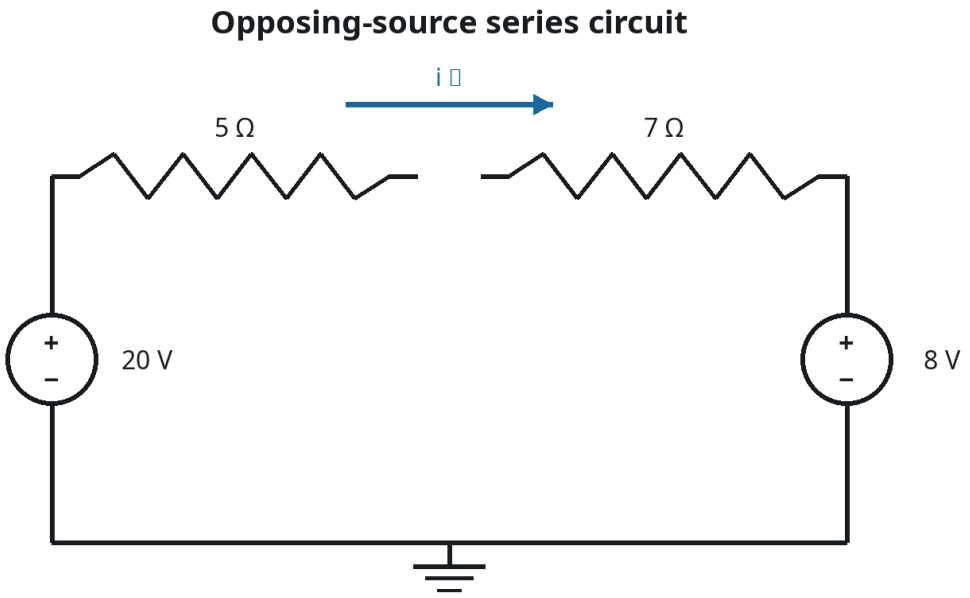
Solution

The source contributions give $v = v_V + v_I = 7.20 + 4.80 = 12.0\,\text{V}$

Power is nonlinear in voltage or current, so the partial powers cannot be added. Use the total response: $P_6 = \frac{v^2}{6} = \frac{(12.0)^2}{6} = 24.0\,\text{W}$ absorbed

Question 3

Use superposition to find the top-branch current, referenced left to right.



A single loop has a 20 V source on the left and an 8 V source on the right, both positive at the top. The top branch contains 5 Ω then 7 Ω; the bottom wire is ground. Current is referenced left to right across the top.

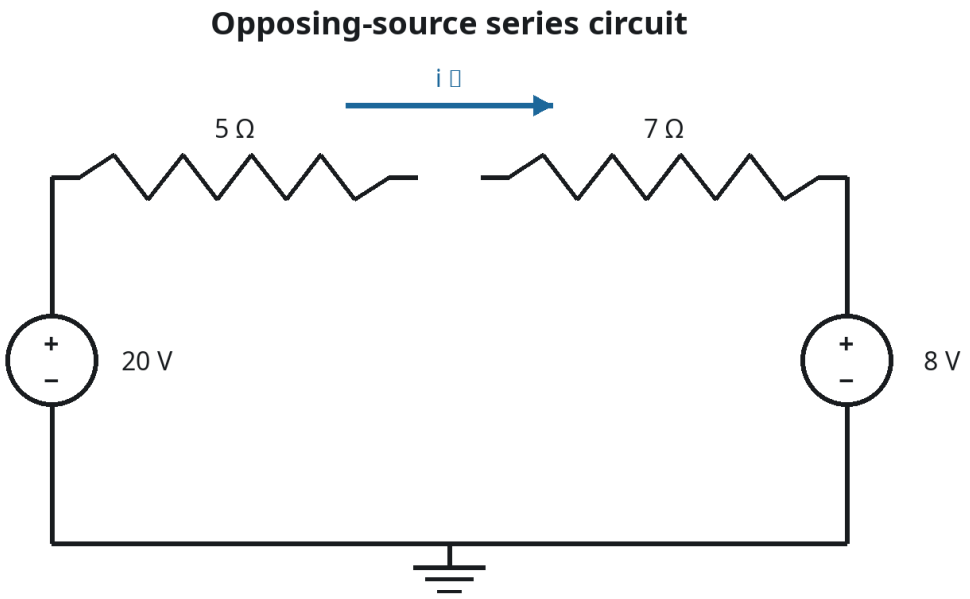
Solution

Define clockwise mesh current i, left to right across the top. With only the $20\,\text{V}$ source active, short the $8\,\text{V}$ source: $i_{20} = \frac{20}{5+7} = 1.6667\,\text{A}$

With only the $8\,\text{V}$ source active, short the $20\,\text{V}$ source. Its polarity drives counterclockwise: $i_8 = -\frac{8}{5+7} = -0.6667\,\text{A}$ $i = i_{20} + i_8 = 1.000\,\text{A}$ left to right

Question 4

Write the governing mesh equation for the full circuit and use it to check the superposition result. State whether the $8\,\text{V}$ source absorbs or delivers power.



A single loop has a 20 V source on the left and an 8 V source on the right, both positive at the top. The top branch contains 5 Ω then 7 Ω; the bottom wire is ground. Current is referenced left to right across the top.

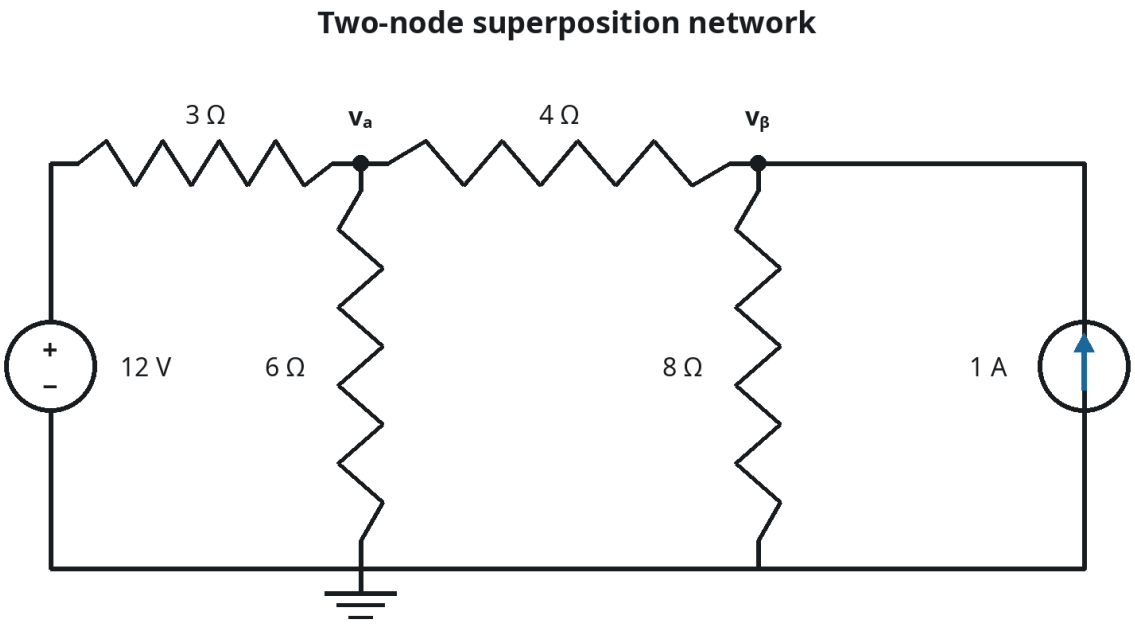
Solution

For clockwise i, KVL is $-20 + 5i + 7i + 8 = 0 \; 12i = 12 \;\Rightarrow\; i = 1.000\,\text{A}$

The clockwise current enters the positive terminal of the right-hand source, so $P_8 = 8i = 8.00\,\text{W}$ absorbed

Question 5

Set up nodal superposition equations and find v_b.



Ground is the bottom wire. A grounded 12 V source feeds va through 3 Ω. A 6 Ω resistor joins va to ground. A 4 Ω resistor joins va to vb. An 8 Ω resistor joins vb to ground, and a 1 A source points upward from ground to vb.

Solution

Take ground as zero. With only the voltage source active,

$$\frac{v_{a,V}-12}{3}+\frac{v_{a,V}}{6}+\frac{v_{a,V}-v_{b,V}}{4}=0$$

$$\frac{v_{b,V}-v_{a,V}}{4}+\frac{v_{b,V}}{8}=0$$

Solving gives $v_{a,V}=6.8571\,\text{V}$, $v_{b,V}=4.5714\,\text{V}$.

With only the current source active, short the voltage source:

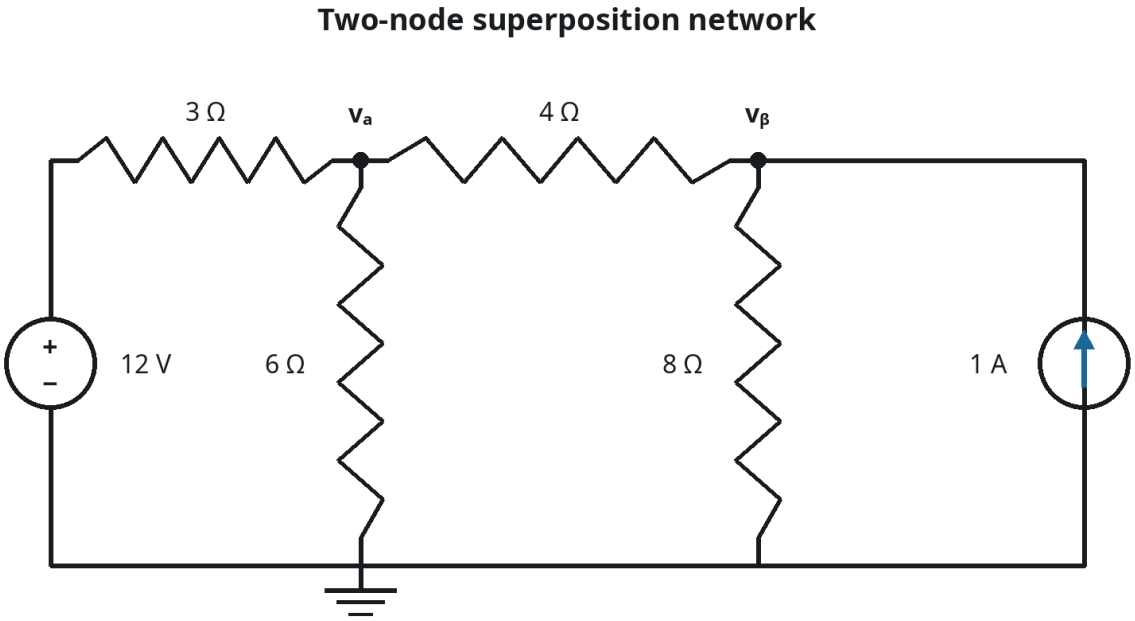
$$\frac{v_{a,I}}{3}+\frac{v_{a,I}}{6}+\frac{v_{a,I}-v_{b,I}}{4}=0$$

$$\frac{v_{b,I}-v_{a,I}}{4}+\frac{v_{b,I}}{8}-1=0$$

Solving gives $v_{a,I}=1.1429\,\text{V}$, $v_{b,I}=3.4286\,\text{V}$. Thus $v_b=4.5714+3.4286=8.000\,\text{V}$.

Question 6

Find the current in the $4\,\Omega$ resistor, positive from v_a to v_b. Interpret a zero result.



Ground is the bottom wire. A grounded 12 V source feeds va through 3 Ω. A 6 Ω resistor joins va to ground. A 4 Ω resistor joins va to vb. An 8 Ω resistor joins vb to ground, and a 1 A source points upward from ground to vb.

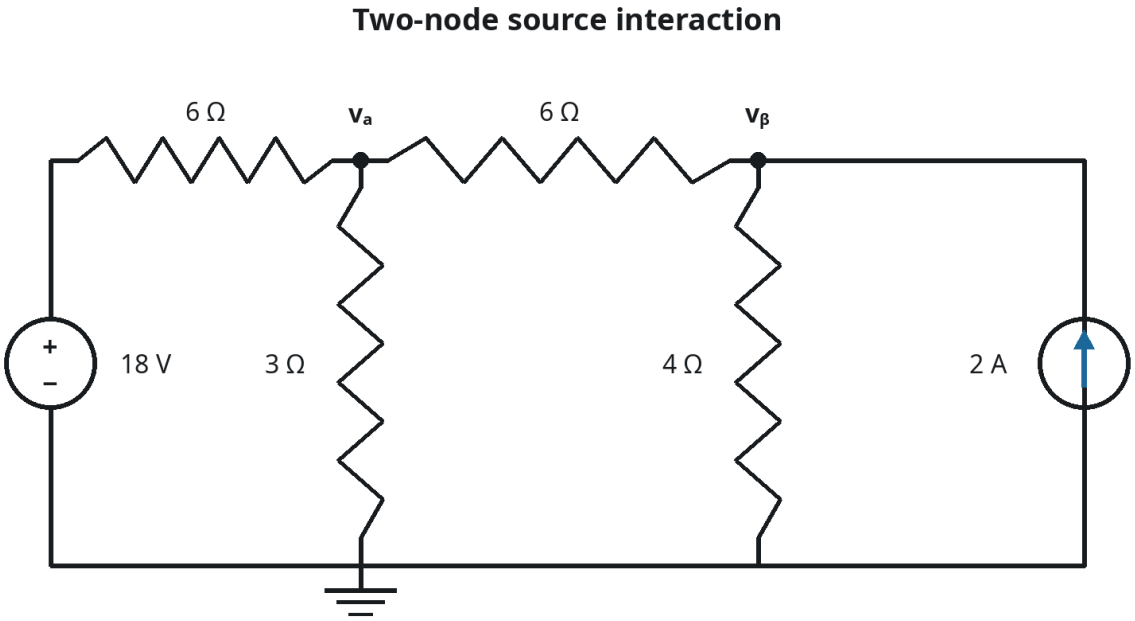
Solution

Using the node contributions, $i_{4,V} = \frac{6.8571-4.5714}{4} = 0.5714\,\text{A}$ $i_{4,I} = \frac{1.1429-3.4286}{4} = -0.5714\,\text{A}$ $i_4 = i_{4,V} + i_{4,I} = 0$

The sources create equal and opposite branch-current contributions. Thus $v_a = v_b = 8.000\,\text{V}$ and no current flows through the $4\,\Omega$ resistor.

Question 7

Use superposition to find v_a and v_b.



Ground is the bottom wire. A grounded 18 V source feeds va through 6 Ω. A 3 Ω resistor joins va to ground. A 6 Ω resistor joins va to vb. A 4 Ω resistor joins vb to ground, and a 2 A source points upward from ground to vb.

Solution

With the current source open,

$$\frac{v_{a,V}-18}{6}+\frac{v_{a,V}}{3}+\frac{v_{a,V}-v_{b,V}}{6}=0,
\qquad \frac{v_{b,V}-v_{a,V}}{6}+\frac{v_{b,V}}{4}=0$$

Solving gives $v_{a,V} = 5.000\,\text{V}, \qquad v_{b,V} = 2.000\,\text{V}$.

With the voltage source shorted,

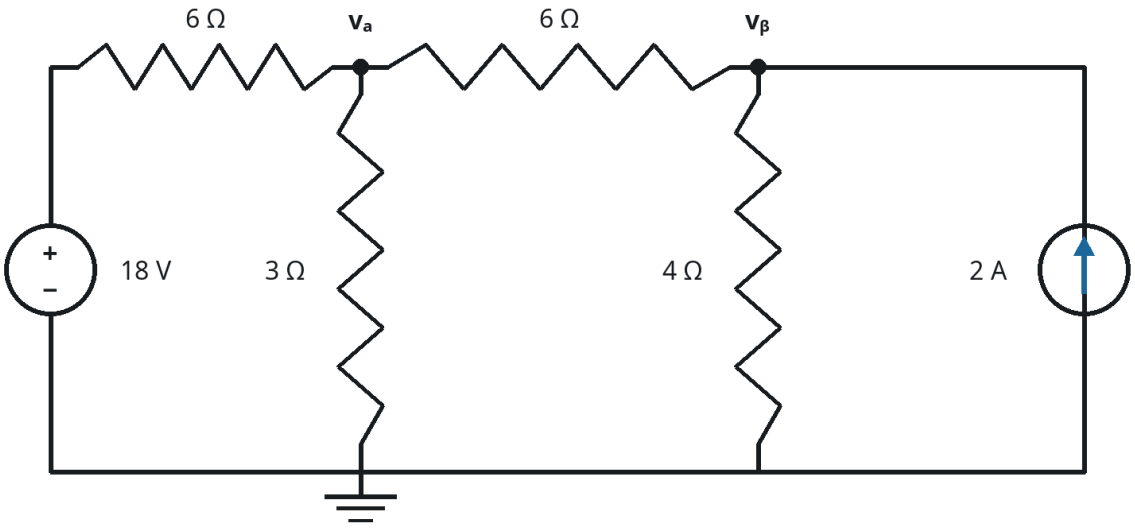
$$\frac{v_{a,I}}{6}+\frac{v_{a,I}}{3}+\frac{v_{a,I}-v_{b,I}}{6}=0,
\qquad \frac{v_{b,I}-v_{a,I}}{6}+\frac{v_{b,I}}{4}-2=0$$

Solving gives $v_{a,I} = 1.333\,\text{V}, \qquad v_{b,I} = 5.333\,\text{V}$. Thus $v_a = 6.333\,\text{V}, \qquad v_b = 7.333\,\text{V}$.

Question 8

Find the current through the resistor between v_a and v_b, positive from v_a to v_b. Which source sets its final direction?

Two-node source interaction



Ground is the bottom wire. A grounded 18 V source feeds va through 6 Ω. A 3 Ω resistor joins va to ground. A 6 Ω resistor joins va to vb. A 4 Ω resistor joins vb to ground, and a 2 A source points upward from ground to vb.

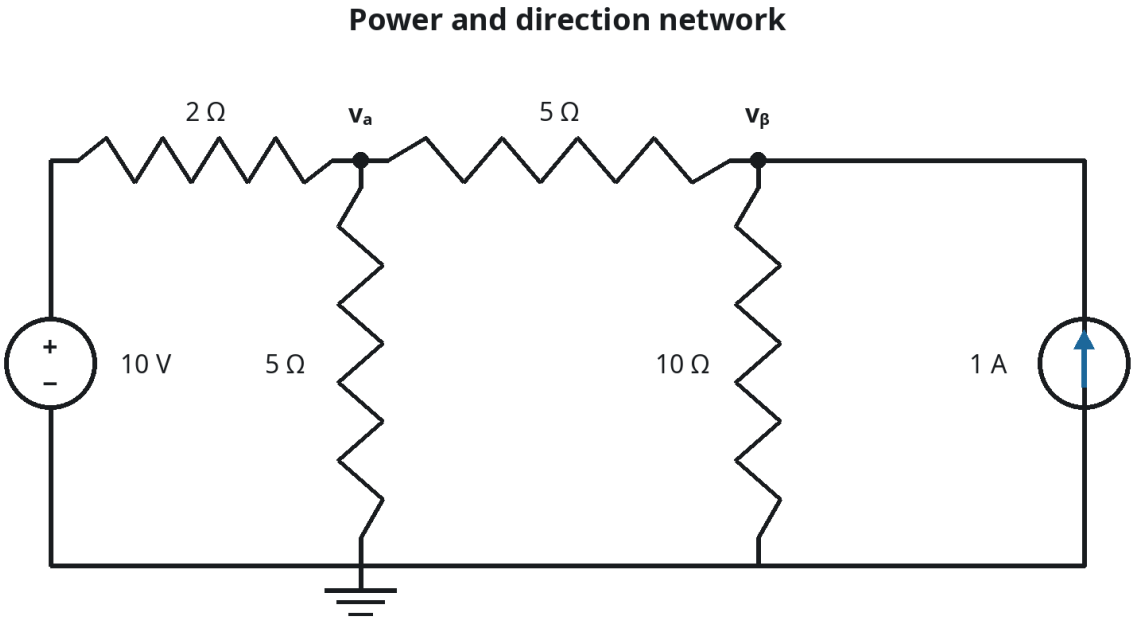
Solution

The partial currents are $i_{6,V} = \frac{5.000-2.000}{6} = 0.500\,\text{A}$ $i_{6,I} = \frac{1.333-5.333}{6} = -0.6667\,\text{A}$ $i_6 = 0.500 - 0.6667 = -0.1667\,\text{A}$

Thus the actual current is $0.1667\,\text{A}$ from v_b to v_a. The current source contribution is larger and sets the final direction.

Question 9

Use superposition to determine the downward current in the $10\,\Omega$ resistor.



Ground is the bottom wire. A grounded 10 V source feeds va through 2 Ω. A 5 Ω shunt joins va to ground. A 5 Ω resistor joins va to vb. A 10 Ω shunt joins vb to ground, and a 1 A source points upward from ground to vb.

Solution

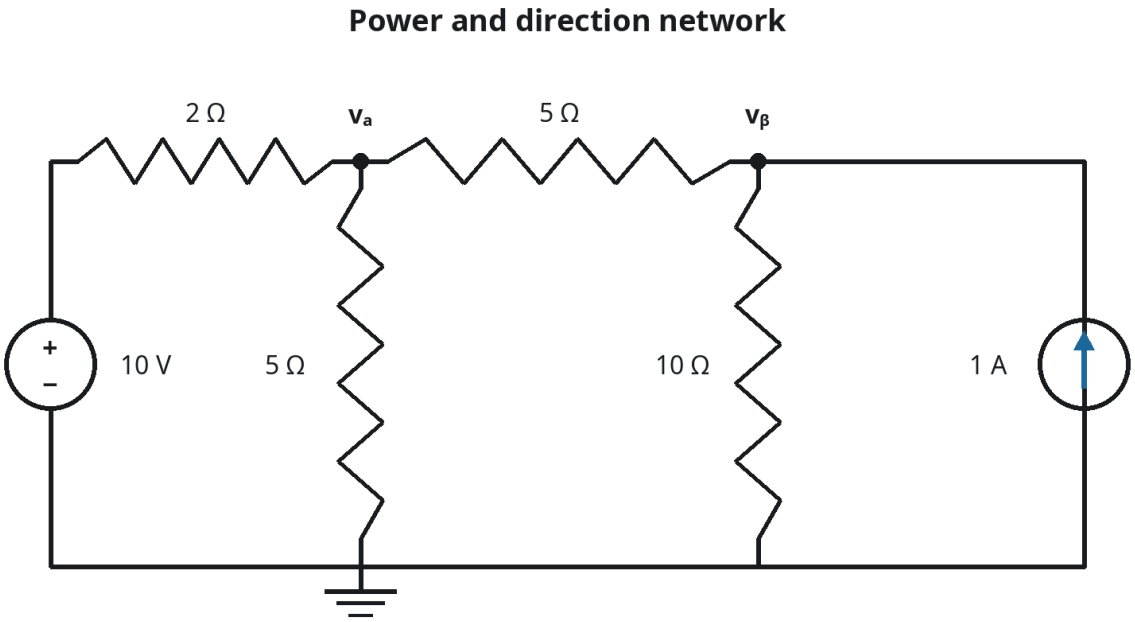
With only the voltage source active, nodal KCL gives $v_{a,V} = 6.5217\,\text{V}, \qquad v_{b,V} = 4.3478\,\text{V}$

With only the current source active, $v_{a,I} = 0.8696\,\text{V}, \qquad v_{b,I} = 3.9130\,\text{V}$

Therefore $i_{10} = \frac{v_{b,V}+v_{b,I}}{10} = \frac{8.2609}{10} = 0.8261\,\text{A}$ downward

Question 10

Find the power absorbed by the $10\,\Omega$ resistor and check total source power against total resistor power.



Ground is the bottom wire. A grounded 10 V source feeds va through 2 Ω. A 5 Ω shunt joins va to ground. A 5 Ω resistor joins va to vb. A 10 Ω shunt joins vb to ground, and a 1 A source points upward from ground to vb.

Solution

The total node voltages are $v_a = 7.3913\,\text{V}, \qquad v_b = 8.2609\,\text{V}$ $P_{10} = \frac{v_b^2}{10} = 6.824\,\text{W}$ absorbed

The resistor powers are

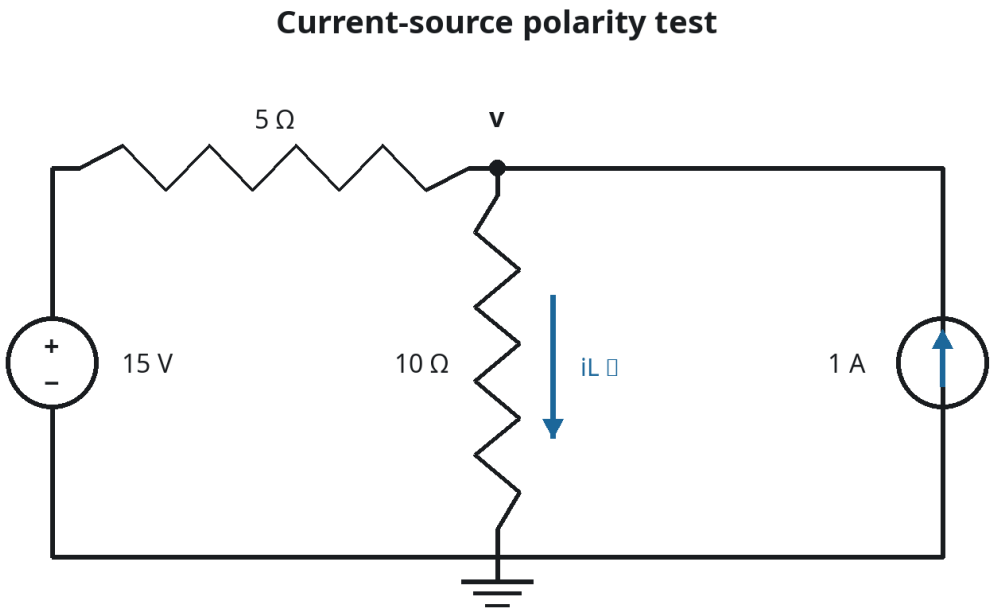
$$P_R = \frac{(10 - v_a)^2}{2} + \frac{v_a^2}{5} + \frac{(v_a - v_b)^2}{5} + \frac{v_b^2}{10} = 21.304\,\text{W}$$

The voltage source supplies current $i_V = (10 - v_a)/2 = 1.30435\,\text{A}$, while the current source has voltage v_b and points upward: $P_{\text{delivered}} = 10(1.30435) + 8.2609(1) = 21.304\,\text{W}$

Thus delivered and absorbed powers agree.

Question 11

Before calculating, state how each independent source is deactivated for superposition. Then find v.



Ground is the bottom wire. A grounded 15 V source feeds node v through 5 Ω. A 10 Ω resistor joins v to ground with downward current. A 1 A source points upward from ground to v.

Solution

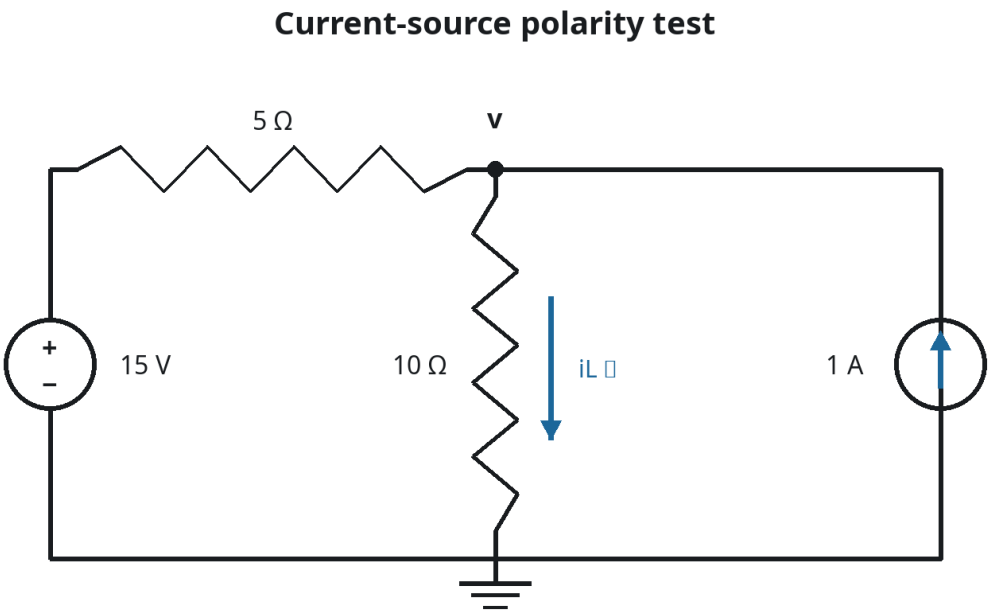
An ideal voltage source is deactivated by replacing it with a short circuit. An ideal current source is deactivated by replacing it with an open circuit.

With only the voltage source active, $v_V = 15\left(\frac{10}{5+10}\right) = 10.0\,\text{V}$

With only the current source active, $v_I = 1(5 \parallel 10) = 3.333\,\text{V}$ $v = v_V + v_I = 13.333\,\text{V}$

Question 12

Reverse only the $1\,\text{A}$ source so it points downward. Use signed contributions to find the new node voltage and state the current direction in the $10\,\Omega$ resistor.



Use the shown circuit except reverse the 1 A source so it points downward from v to ground. The 15 V source feeds v through 5 Ω, and 10 Ω joins v to ground.

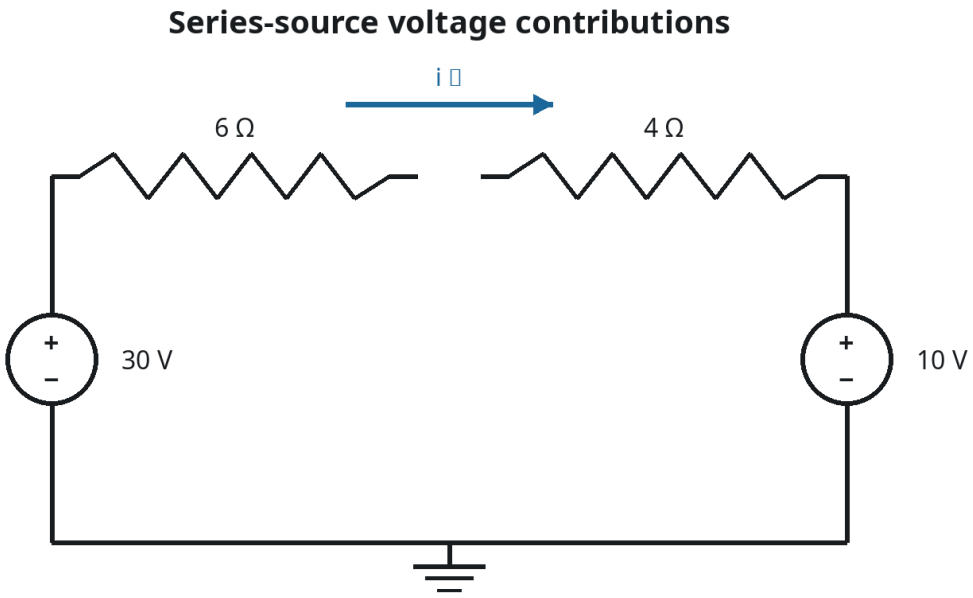
Solution

Keep downward load current positive. The voltage-source contribution is unchanged: $v_V = 10.0\,\text{V}$

Reversing the current source reverses its contribution: $v_I = -1(5 \parallel 10) = -3.333\,\text{V}$ $v = 10.0 - 3.333 = 6.667\,\text{V}$ $i_{10} = \frac{v}{10} = 0.6667\,\text{A}$ downward

Question 13

Find the loop current by superposition and give both signed source contributions.



A single loop has a 30 V source on the left and a 10 V source on the right, both positive at the top. The top branch contains 6 Ω then 4 Ω. The bottom wire is ground, and current is referenced left to right across the top.

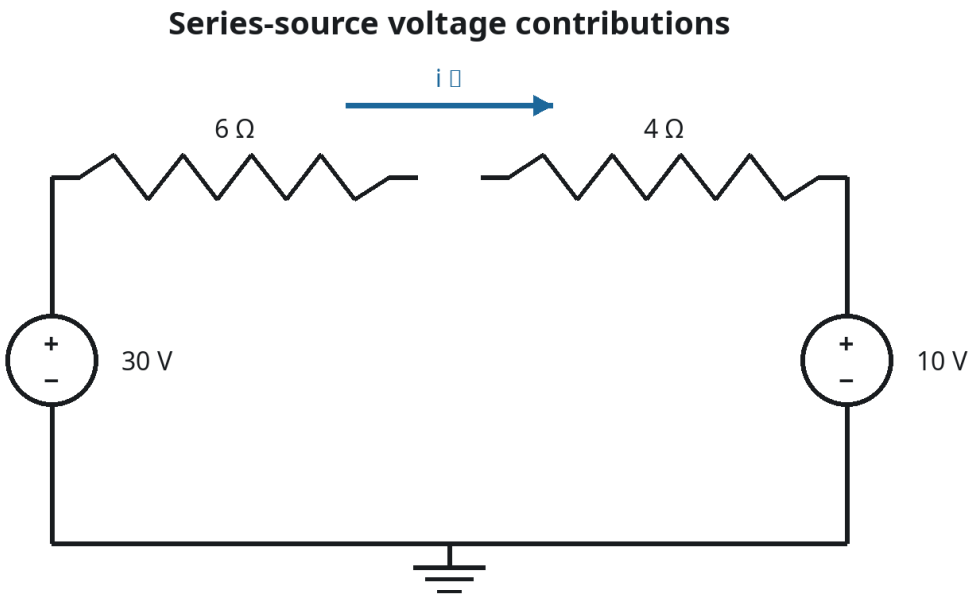
Solution

Define clockwise current i, left to right on the top branch. $i_{30} = \frac{30}{6+4} = 3.00\,\text{A}$

The right source opposes the reference direction: $i_{10} = -\frac{10}{6+4} = -1.00\,\text{A}$ $i = i_{30} + i_{10} = 2.00\,\text{A}$ left to right

Question 14

Find the voltage across the $4\,\Omega$ resistor, positive on its left terminal, and the power delivered or absorbed by each source.



A single loop has a 30 V source on the left and a 10 V source on the right, both positive at the top. The top branch contains 6 Ω then 4 Ω. The bottom wire is ground, and current is referenced left to right across the top.

Solution

From the superposed current, $v_4 = 4i = 4(2.00) = 8.00\,\text{V}$

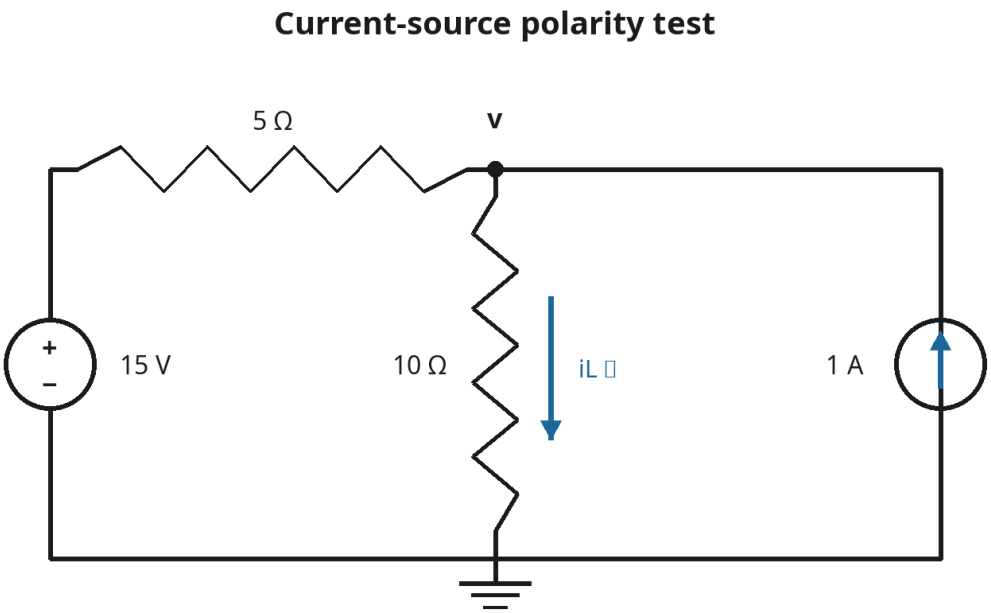
Current leaves the positive terminal of the $30\,\text{V}$ source, so it delivers $P_{30} = 30(2.00) = 60.0\,\text{W}$ delivered

Current enters the positive terminal of the $10\,\text{V}$ source, so it absorbs $P_{10} = 10(2.00) = 20.0\,\text{W}$ absorbed

The resistors absorb $i^2(6+4) = 40.0\,\text{W}$, which checks the balance.

Question 15

Add a dependent current source of value $0.20v$ A pointing upward from ground to node v in parallel with the shown elements. Use superposition of the independent sources to find v. Keep the dependent source active in every subproblem.



Ground is the bottom wire. A grounded 15 V source feeds node v through 5 Ω. A 10 Ω resistor joins v to ground. Independent 1 A and added dependent 0.20v A sources both point upward from ground to v.

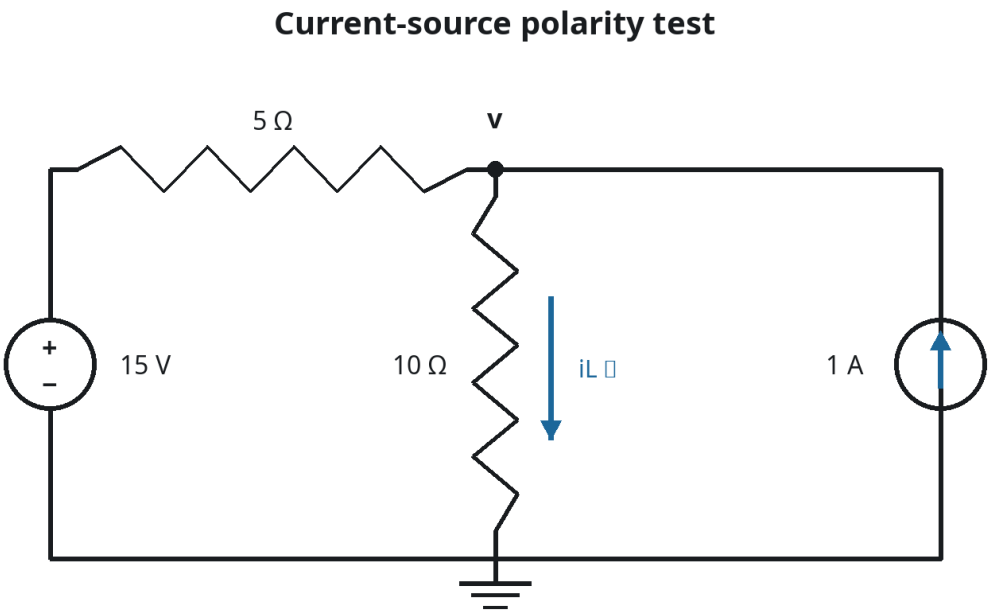
Solution

Dependent sources are never deactivated. With only the $15\,\text{V}$ source active, $\frac{v_V-15}{5}+\frac{v_V}{10}-0.20v_V=0$ $0.10v_V=3 \;\Rightarrow\; v_V=30.0\,\text{V}$

With only the independent $1\,\text{A}$ source active, $\frac{v_I}{5}+\frac{v_I}{10}-0.20v_I-1=0$ $0.10v_I=1 \;\Rightarrow\; v_I=10.0\,\text{V}$ $v=v_V+v_I=40.0\,\text{V}$

Question 16

For the modified circuit in Question 15, check the result using one full-circuit KCL equation and find the dependent-source current.



Use the Question 15 circuit: the 15 V source feeds v through 5 Ω; 10 Ω joins v to ground; independent 1 A and dependent 0.20v A sources point upward from ground to v.

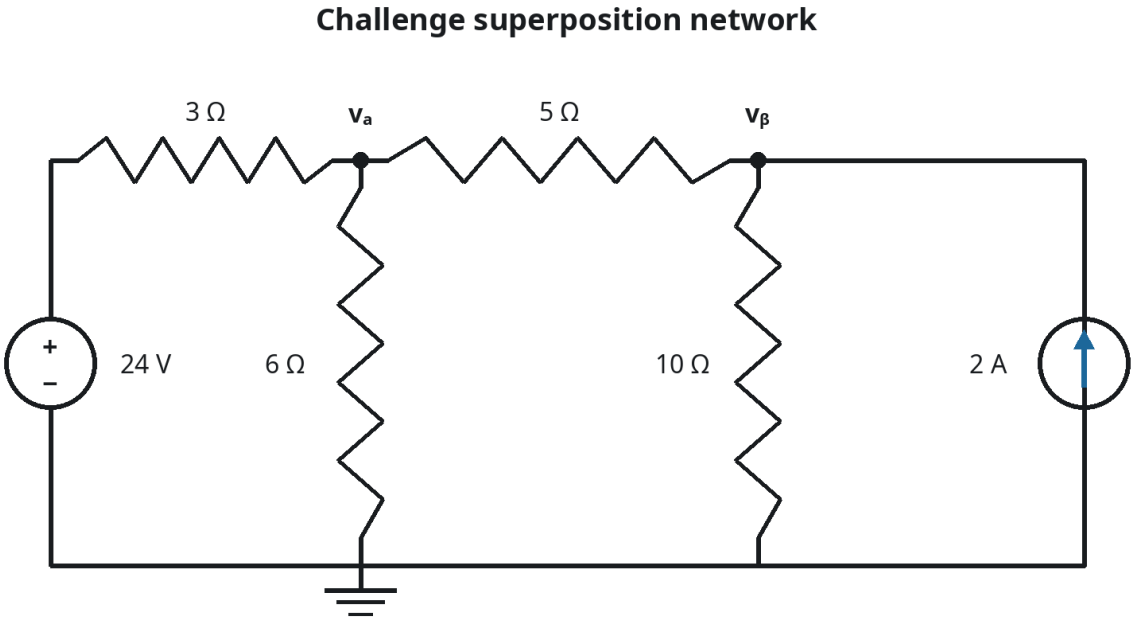
Solution

KCL at v for the full circuit is $\frac{v-15}{5} + \frac{v}{10} - 1 - 0.20v = 0$ $0.10v - 4 = 0 \;\Rightarrow\; v = 40.0\,\text{V}$

The controlled current is $i_d = 0.20v = 0.20(40.0) = 8.00\,\text{A}$ upward

Question 17

Use superposition to find v_a and v_b for this challenge network.



Ground is the bottom wire. A grounded 24 V source feeds va through 3 Ω. A 6 Ω resistor joins va to ground. A 5 Ω resistor joins va to vb. A 10 Ω resistor joins vb to ground, and a 2 A source points upward from ground to vb.

Solution

With the current source open,

$$\frac{v_{a,V}-24}{3}+\frac{v_{a,V}}{6}+\frac{v_{a,V}-v_{b,V}}{5}=0,
\qquad \frac{v_{b,V}-v_{a,V}}{5}+\frac{v_{b,V}}{10}=0$$

Solving gives $v_{a,V} = 14.1176\,\text{V}, \qquad v_{b,V} = 9.4118\,\text{V}$.

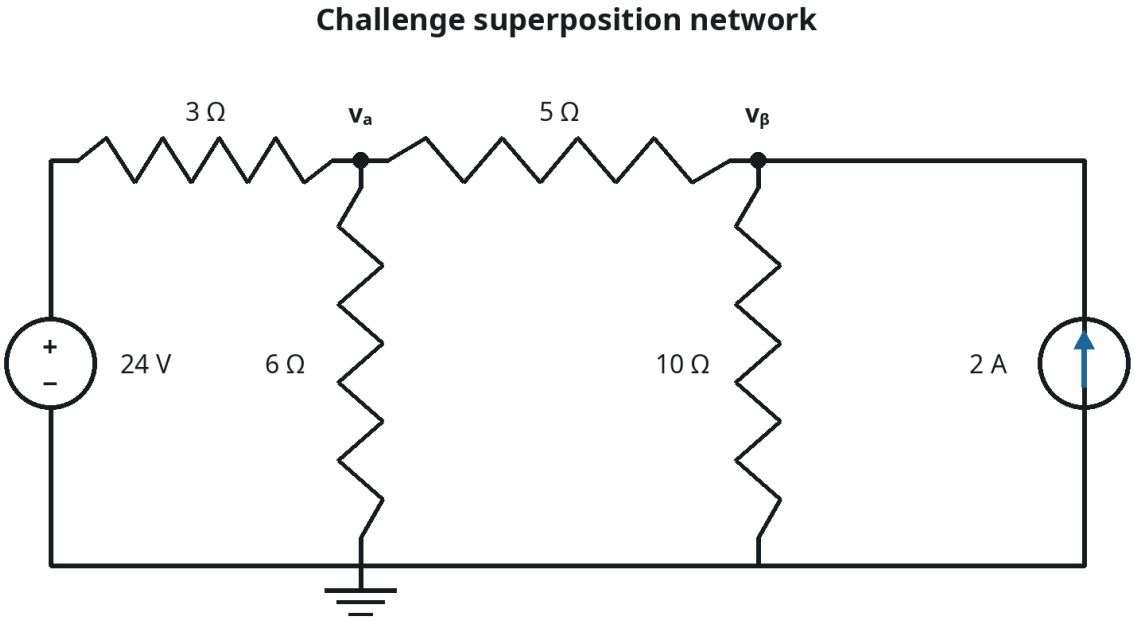
With the voltage source shorted,

$$\frac{v_{a,I}}{3}+\frac{v_{a,I}}{6}+\frac{v_{a,I}-v_{b,I}}{5}=0,
\qquad \frac{v_{b,I}-v_{a,I}}{5}+\frac{v_{b,I}}{10}-2=0$$

Solving gives $v_{a,I} = 2.3529\,\text{V}, \qquad v_{b,I} = 8.2353\,\text{V}$. Thus $v_a = 16.4706\,\text{V}, \qquad v_b = 17.6471\,\text{V}$.

Question 18

Find the current in the $5\,\Omega$ resistor, positive from v_a to v_b, and interpret its sign.



Ground is the bottom wire. A grounded 24 V source feeds va through 3 Ω. A 6 Ω resistor joins va to ground. A 5 Ω resistor joins va to vb. A 10 Ω resistor joins vb to ground, and a 2 A source points upward from ground to vb.

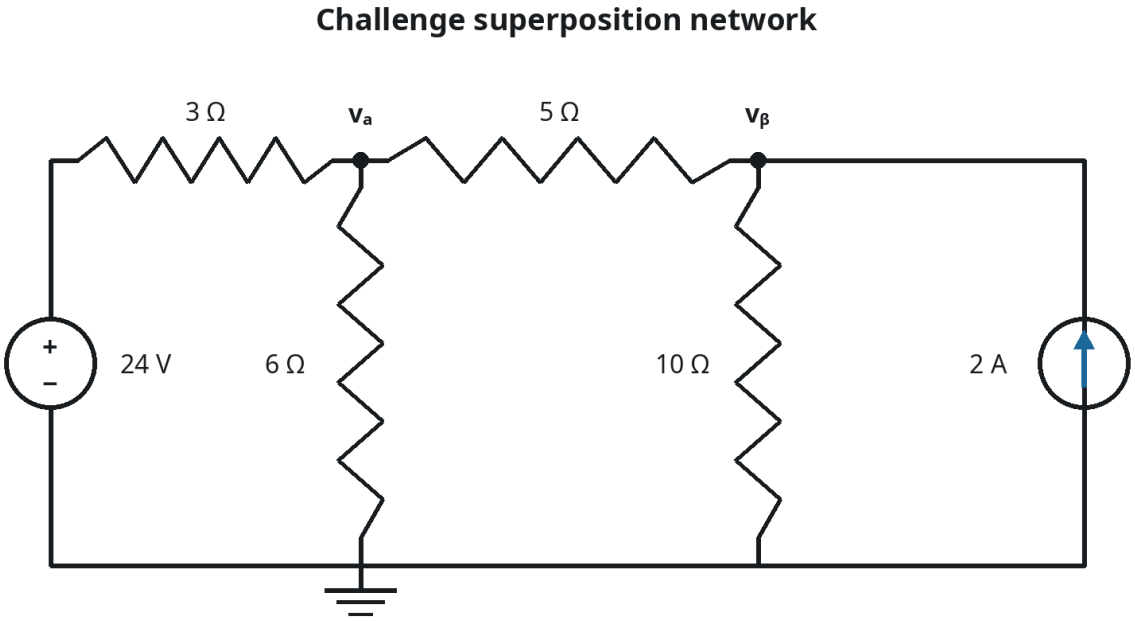
Solution

The source contributions are $i_{5,V} = \frac{14.1176-9.4118}{5} = 0.9412\,\text{A}$ $i_{5,I} = \frac{2.3529-8.2353}{5} = -1.1765\,\text{A}$ $i_5 = 0.9412 - 1.1765 = -0.2353\,\text{A}$

The negative sign means the actual current is $0.2353\,\text{A}$ from v_b to v_a.

Question 19

Find the power absorbed by the $10\,\Omega$ resistor. Show why adding partial powers would be wrong.



Ground is the bottom wire. A grounded 24 V source feeds va through 3 Ω. A 6 Ω resistor joins va to ground. A 5 Ω resistor joins va to vb. A 10 Ω resistor joins vb to ground, and a 2 A source points upward from ground to vb.

Solution

First add the voltage contributions: $v_b = 9.4118 + 8.2353 = 17.6471\,\text{V}$

Then calculate power from the total response: $P_{10} = \frac{v_b^2}{10} = 31.14\,\text{W}$ absorbed

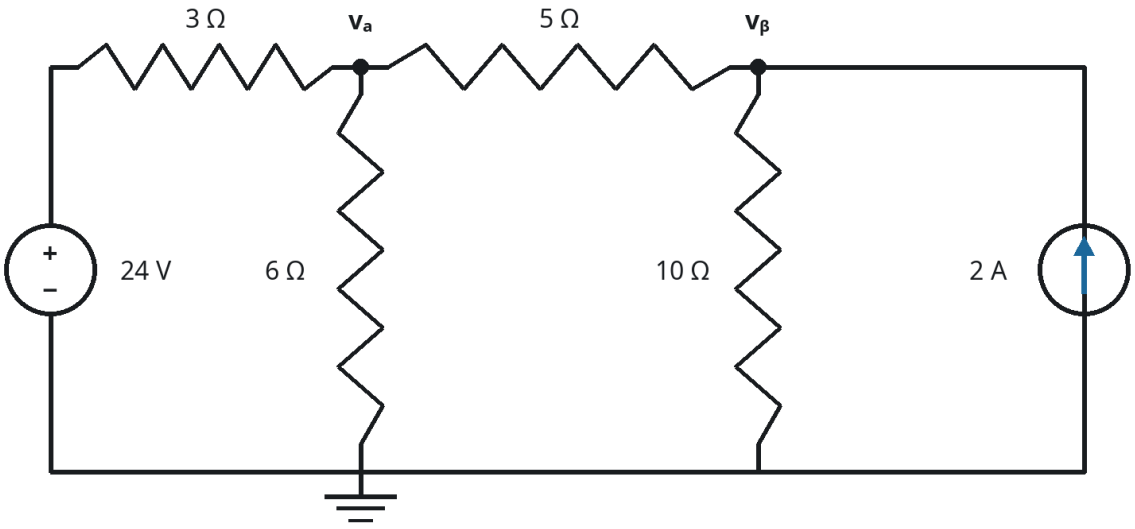
Adding partial powers would give $\frac{(9.4118)^2}{10} + \frac{(8.2353)^2}{10} = 15.64\,\text{W} \neq 31.14\,\text{W}$

The missing cross term shows that power does not obey superposition.

Question 20

Perform a full-circuit power check. State the absorbed or delivered status of both independent sources.

Challenge superposition network



Ground is the bottom wire. A grounded 24 V source feeds va through 3 Ω. A 6 Ω resistor joins va to ground. A 5 Ω resistor joins va to vb. A 10 Ω resistor joins vb to ground, and a 2 A source points upward from ground to vb.

Solution

Using $v_a = 16.4706\,\text{V}$ and $v_b = 17.6471\,\text{V}$, total resistor absorption is

$$P_R = \frac{(24 - v_a)^2}{3} + \frac{v_a^2}{6} + \frac{(v_a - v_b)^2}{5} + \frac{v_b^2}{10} = 95.53\,\text{W}$$

The $24\,\text{V}$ source current leaving its positive terminal is $i_{24} = \frac{24 - v_a}{3} = 2.5098\,\text{A}$ $P_{24} = 24 i_{24} = 60.24\,\text{W}$ delivered

The $2\,\text{A}$ source current leaves its positive top terminal, so $P_{2A} = v_b(2) = 35.29\,\text{W}$ delivered $P_{\text{delivered}} = 60.24 + 35.29 = 95.53\,\text{W}$

Thus $P_R = P_{\text{delivered}} = 95.53\,\text{W}$ within rounding. Both sources deliver power.